

Open Design Project: Client Outline and Pitch

Team: Spotted... then Stopped

Clients: Cornell CALS Extension / E&J Gallo Winery / National Grape

Problem Statement:

New York vineyard owners are trying to maximize yield in their vineyards, but Spotted Lanternflies are infesting vines and contaminating harvests, leading to decreased yield. One of the most thought of solutions would be to shake the plant enough for the bugs to fall, but there is the concern that the grapes would be lost as well. Although insecticides are normally used to control pests, this is not the most desirable solution in this case as they are both expensive and require continuous use, thus not as useful in the long term. Another challenge is that lanternflies killed while on the vines may fall into the harvester and get mixed into the produce, which cannot be filtered due to concerns for texture and flavor. A joint study conducted by Cornell and Penn State found that 60% of lantern flies on vines wound up in the harvest bins. Current health standards require the product to be rejected if even 1-2 adults are found in a sample of 1 kilogram.

Impact:

Reducing contamination during the harvesting process leads to less wasted product and maximizes yield.

Proposed Direction:

Our main idea is to have a machine attached to the front of the harvester that would get rid of the flies right before the grapes are collected.

Concept: Spinning Brushes

An attachment to the front of the vehicle that has rotating brushes aimed at only the vine. The brushes would be located on a rotating rod (almost like a car wash brush) so as to scrape the flies off the vine without disturbing the grapes too much.

How it would be used:

- Attach rotating rods to the harvester at correct height
- Turn them on to spin
- Drive the harvester through to collect grapes

Why it's better than the status quo:

- Flies would be knocked off vines before reaching harvester
- Brushes below grapes to protect harvest
- Don't need to worry about separating grapes vs flies

End-of-semester proof-of-concept:

- A large, slowly spinning, brush attached to a rod.

Key Risks / Unknowns:

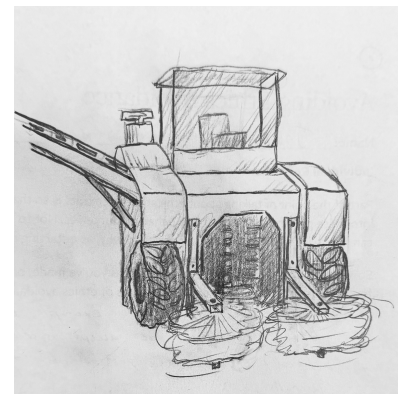
Damage to vines — Could reduce yield in the long run if the vine is scraped: We could test the bristles on a vine and see how it's damaged.

Knocking off grapes prematurely — This could get rid of some of the harvest that we want: We could test to make sure to find the right height to direct the bristles at to avoid hitting grapes.

Gaps between brushes may leave some flies on vines — This might leave some contamination still: We could test this by running a prototype on a model of the vine with attachments of similar grip strength, and seeing what is not removed.

Questions for the client

1. Can we get a blueprint/design for the harvesters used?



Affects how the brush would connect to the harvester, and the design considerations that come with that.

2. What is the average height/length of the vine?

Affects the dimensions of the brushes needed.

3. What is the durability of the vine?

Affects the amount of force we can put on the vine and the material used for the bristles.

References:

<https://wineamerica.org/wp-content/uploads/2025/05/2025-WineAmerica-Methodology-4-28-25-Final.pdf>